

FASD | Digital library

Overview

Goals

1. Create an ER diagram based on the schema below
2. Create API endpoints as per specified controllers
3. Adding Spring Security [HOLD]
4. Create React based Frontend application for this project

Specifications

Entities:

1. Books
2. Author
3. User
4. Student
5. Card
6. Transaction

Database Schema

Author

1. Id
2. Name
3. Email
4. Age
5. Country

Book

1. Id
2. Name
3. Author_ID
4. Number of Pages
5. Language
6. Available
7. Genre
8. ISBN Number
9. Published Date

User

1. Id
2. Name
3. Username
4. Password
5. Email
6. Authority/Role

Student

1. Id
2. Age
3. Name
4. Country
5. Email

6. Phone Number
7. CreatedOn
8. UpdatedOn
9. CardId

Card

1. Id
2. Status
3. Email
4. ValidUpto
5. CreatedOn
6. UpdatedOn

Transaction

1. Id
2. CardId
3. BookId
4. Date
5. BookDueDate
6. Isissued
7. IsReturned
8. FineAmount
9. Status
10. CreatedOn
11. UpdatedOn

Entity Relation:

Source	Destination	Relation
Student	Card	1-1
Book	Author	1-1
Book	Transaction	1-N
Card	Transaction	1-N

Controllers:

StudentController

Mapping: /student/<endpoint>

1. CRUD API

BookController

Mapping: /book/<endpoint>

1. CRUD API for Book

AuthorController

Mapping: /author/<endpoint>

1. CRUD API

TransactionController

Mapping: /transact/<endpoint>

1. IssueBook (Param: BookID,CardID)
 1. If Card is active
 2. If Book is available
 3. Each card can have utmost 3 books
 4. If all of the above is ok, then issue the book:

1. Create a new transaction
 2. Mark the book as unavailable
5. If it fails, return an error and update the status as Failure
2. Return (Param: BookId, CardId)
 1. Check if booked and carded are valid combination
 2. If Card is active
 3. If the above is ok, then process the return
 1. Mark the book as available
 2. Each card can have utmost 3 books
 3. If today's date is higher than due date, calculate the fine and store it
 4. In transaction table, set card_id to null
3. Get details of a transaction id
4. Get all transactions of book_id
4. Get all transactions of card_id